Multi-Step Word Problems Without Parentheses

Multi-Step Word Problems • Standard: CCSS.MATH.4.OA.A.3

BOOK STORY OPENER: Inside the 4-Floor Number Elevator (T, Z, Y, X) & Young Gauss's 10-Second Shortcut

In 1786, 9-year-old Carl Friedrich Gauss stunned his teacher by adding 1 through 100 in seconds—folding the list into 50 pairs of 101! In Chapters 2 & 4 of From Blocks to the Stack, students discover how the 4-Floor Number Elevator (Shelves X, Y, Z, T) automatically lifts finished answers up to Shelf Y when you type the next number, solving multi-step stories with zero parentheses.

Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 2: The Number Elevator & Ch. 4: Young Gauss's 10-Second Shortcut

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Place Value Blocks & 4-Register Stack

Use Place Value unit blocks to represent quantities alongside the 4-line stack display.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

3. CALCULATOR ACTION (SC-6 KEYCAPS & STACK TRACE)

Keystroke Flow: [8] [ENTER] [3] [-] [2] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 × 2 = 10 in X!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Unlike traditional algebraic calculators that require parenthesis keys, RPN evaluates inner expressions first, lifting results naturally.

Core Investigation Problem:

A baker has 15 cookies, sells 6, and packs the rest into 3 bags. How many per bag? Keystrokes for (15 - 6) / 3.

Automatic Stack Lift & Register Retention

Order of Operations & Stack Lift • Standard: CCSS.MATH.4.OA.A.3

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Observe the 4-line stack screen as values automatically lift into register Y: (12 / 4) + 7. Prediction: 12 / 4 = 3. Entering 7 will lift 3 to Y. 3 + 7 = 10.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[1] [2] [ENTER] [4] [÷] [7] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	First digit of 12 enters X
[2]	12	12 in register X
[ENTER]	12	12 copied to Y
[4]	4	4 enters X
[÷]	3	12 / 4 = 3 in X
[7]	7	7 entered: 3 lifts to Y automatically!
[+]	10	3 + 7 = 10!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [ENTER] after an operator (+, −, ×, ÷) before typing the next number in a chained calculation, wasting a stack register.

Hands-On Block Remediation: Demonstrate Automatic Stack Lift: after [18] [ENTER] [6] [−] produces 12, typing [3] automatically lifts 12 to Register Y—no [ENTER] needed before [3] [×]!

Misconception 2: Reversing operand order in subtraction or division (computing X − Y instead of Y − X).

Hands-On Block Remediation: Remind students that Register Y is always the starting quantity on top and Register X is the operator's action; if entered backward, press [x swap y] before [−] or [÷].

4. EXTENSION MATHEMATICAL INSIGHT

Automatic lift prevents unnecessary ENTER presses between sequential chained operations.

The 4-Register Stack: Four Operations & Automatic Lift

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: 4-REGISTER STACK ANATOMY

- Four registers: T (top), Z (third), Y (second/operand), X (screen/bottom).
- ENTER duplicates X into Y and pushes older values up into Z and T.
- Two-number operators (+, -, ×, ÷) combine Y and X, dropping Z to Y and T to Z.

Key Sequence: `8` `ENTER` `3` `-` `2` `×`

Problem 1: Calculate (14 - 5) × 3 on the stack.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 2: Calculate 30 / (2 + 3): add denominator first!

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 3: Calculate (9 + 7) / (8 - 4) using 4 registers.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 4: A farmer collects 20 eggs, breaks 2, and divides among 6 cartons.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Unit 05: The 4-Register Stack: Four Operations & Automatic Lift — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 3–4

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: A baker has 15 cookies, sells 6, and packs the rest into 3 bags. How many per bag? Keystrokes for (15 - 6) / 3.

Core Solution: 3 cookies per bag. Keystrokes: [1][5] [ENTER] [6] [-] [3] [÷]

Core Check Keys: 1 5 ENTER 6 - 3 ÷

Extension Challenge: Calculate (24 / 6) + 15 on the calculator without pressing ENTER twice.

Extension Solution: 19. Keystrokes: [2][4] [ENTER] [6] [÷] [1][5] [+]

Ext. Check Keys: 2 4 ENTER 6 ÷ 1 5 +

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Calculate (14 - 5) × 3 on the stack.

Expected Answer: 27

SC-6 Verification Keys: 1 4 ENTER 5 - 3 ×

Problem 2: Calculate 30 / (2 + 3): add denominator first!

Expected Answer: 6

SC-6 Verification Keys: 2 ENTER 3 + 3 0 ↑↓ ÷

Problem 3: Calculate (9 + 7) / (8 - 4) using 4 registers.

Expected Answer: 4

SC-6 Verification Keys: 9 ENTER 7 + 8 ENTER 4 - ÷

Problem 4: A farmer collects 20 eggs, breaks 2, and divides among 6 cartons.

Expected Answer: 3 eggs

SC-6 Verification Keys: 2 0 ENTER 2 - 6 ÷

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.